

**Beyond Accuracy: A Robustness Audit of Health Misinformation
Classification Across CoAID and FakeHealth**

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Introduction and Problem

Health misinformation creates a management problem as well as a technical one. Organizations that monitor public communication must decide which items deserve attention, what evidence supports a warning, and when an automated system should defer to a person. A high classification score is insufficient if a model learns publisher names, pandemic vocabulary, or dataset formatting instead of transferable indicators of information quality. Systematic reviews likewise show that health misinformation varies across topics, platforms, definitions, and data-collection procedures (Baris Schlicht et al., 2023; Suarez-Lledo & Alvarez-Galvez, 2021). The practical question is therefore not simply whether unreliable content can be classified, but whether the resulting decision rule remains credible under changes in source, time, dataset, and label construction.

This study develops a reproducible audit of natural-language-processing models using CoAID and FakeHealth. CoAID contains COVID-19 news and claims labeled through the dataset’s credibility process (Cui & Lee, 2020). FakeHealth combines health stories and news releases with HealthNewsReview ratings, review explanations, and social-engagement metadata (Dai et al., 2020). These resources are complementary rather than interchangeable: one is pandemic-specific and strongly source-structured, whereas the other covers broader health reporting and uses expert-review criteria. Throughout this report, *unreliable* refers to the harmonized dataset label, not an independent medical diagnosis.

Five questions guide the analysis: How well do sparse and transformer text models classify each dataset? Do results survive source, temporal, and artifact controls? Do models transfer across datasets? Does engagement add information beyond text? Can calibration, narrative analysis, explanations, and structured review identify cases that should not be trusted automatically? The contribution is an integrated robustness and decision-support evaluation rather than a new classification architecture. Negative findings are consequently informative: a loss under source control or transfer demonstrates a boundary on what the learned model can support.

Background

Prior research establishes both the public-health relevance of misinformation and the difficulty of defining it consistently. Its spread depends on messages, actors, networks, and interpreters (Wang et al., 2019). Detection studies use labels ranging from expert judgments to publisher proxies, so internal accuracy cannot substitute for external validity (Baris Schlicht et al., 2023). This study therefore pairs BERT-style contextual encoders (Devlin et al., 2019) with interpretable TF-IDF controls, calibration, and human review.

Data and Method

The acquisition process pinned official revisions and retained inventories, checksums, native labels, parsing outcomes, duplicates, and exclusions. Of 8,271 parsed records, the primary cohort contains 3,055 CoAID articles and 2,137 FakeHealth items. CoAID claims lack comparable article bodies and were excluded. Other coded exclusions cover unavailable or placeholder pages, duplicates, conflicting labels, and bodies under 30 words; excluded rows remain auditable.

Standard manifests use approximately 70% training, 15% validation, and 15% testing while keeping connected duplicate families in one partition. Controlled evaluation follows the strongest feasible design. FakeHealth uses a canonical-publisher-disjoint split, ten repeated valid publisher allocations, and a hosting-domain sensitivity test. CoAID cannot satisfy the frozen publisher-disjoint feasibility gates because its 233 unreliable articles are concentrated among few sources; it therefore uses a temporal, family-disjoint fallback. These controlled conditions answer related but nonidentical questions and are not pooled.

Sparse candidates include word/bigram TF-IDF Logistic Regression and Linear SVM, publisher-, unit-, and length-only diagnostics, title/body ablations, and artifact-masked text. Each eligible sparse model compares 24 configurations on validation macro F1 before one test evaluation. Transformer benchmarks include pinned DistilBERT, ModernBERT, and BiomedBERT checkpoints, three seeds, early stopping, and a validation-only learning-rate audit. All were executed on Apple MPS. Transformer inputs are truncated to 512 tokens: only about 1.5% of CoAID exceeds this limit, compared with 81.8% of FakeHealth. The latter results therefore describe leading-window classification, not full-article transformer analysis.

Macro F1 is primary because both classes matter despite imbalance; unreliable-class precision, recall, F1, and precision-recall area are retained. Test uncertainty uses 1,000 family-, publisher-, or domain-grouped bootstrap resamples, as appropriate. Paired resampling compares models on identical records. Calibration methods and selective-prediction thresholds are chosen using separated validation subsets before test evaluation, following the principle that confidence should correspond to observed correctness (Guo et al., 2017). Transfer models train on one dataset and score the other’s held-out test records after removing target domains seen during source training or validation.

FakeHealth engagement experiments compare text, engagement-ID counts, and combined models on the same complete cases. BERTopic clusters unreliable-labeled records; representatives are reviewed before a cluster is called a narrative (Grootendorst, 2022). A frozen 80-case audit samples ordinary and high-confidence errors, deferred cases, transfer failures, and narrative cases. Blind Stage A precedes explanations; Stage B confirms proposed error codes.

Findings: Classification and Robustness

Figure 1 compares the primary sparse baseline with the strongest transformer point estimate in each declared condition. Standard performance is high on CoAID, but the apparent strength is not source-independent. Logistic Regression falls from .857 macro F1 on the standard split to .675 temporally, while publisher-only Logistic Regression reaches .978 standard and .938 temporal. CoAID’s language models therefore learn useful discrimination within the corpus, yet publisher identity remains a stronger shortcut than full text. The temporal transformer scores do not remove this concern because publishers still overlap across time.



Figure 1

Primary macro-F1 results across standard and controlled tests. Transformer bars show the highest three-seed architecture mean within each condition, not a single universally selected model.

FakeHealth provides the stronger source-control result. Logistic Regression scores .677 standard and .660 with publishers held out; across ten valid publisher allocations its mean is .651 (SD = .062). The hosting-domain score is .516, but its 95% grouped-bootstrap interval (.266, .774) reflects only about five effective domain groups. Transformers do not clearly exceed the sparse baseline on ordinary FakeHealth evaluation: ModernBERT’s best standard mean is .674. BiomedBERT reaches .628 in the domain test, yet architecture rank changes across conditions and the test remains small. The binary treatment of FakeHealth rating 3 is also consequential: excluding boundary-rating training items raises same-record Logistic Regression macro F1 from .735 to .791. Collectively, these results favor robustness profiles over a single leaderboard score.

Findings: Transfer, Decision Support, and Interpretation

Cross-dataset transfer supplies the clearest external-validity test. As Figure 2 shows, full-text Logistic Regression trained on CoAID scores .421 on 263 filtered FakeHealth test records, compared with .686 for the target-trained same-record baseline. Training on FakeHealth and transferring to 161 CoAID records is worse: .309 versus .874. These losses reject portability of the fitted decision rules. They do not establish that either dataset is medically correct; topic, publication period, text unit, publisher composition, and label provenance all change simultaneously.

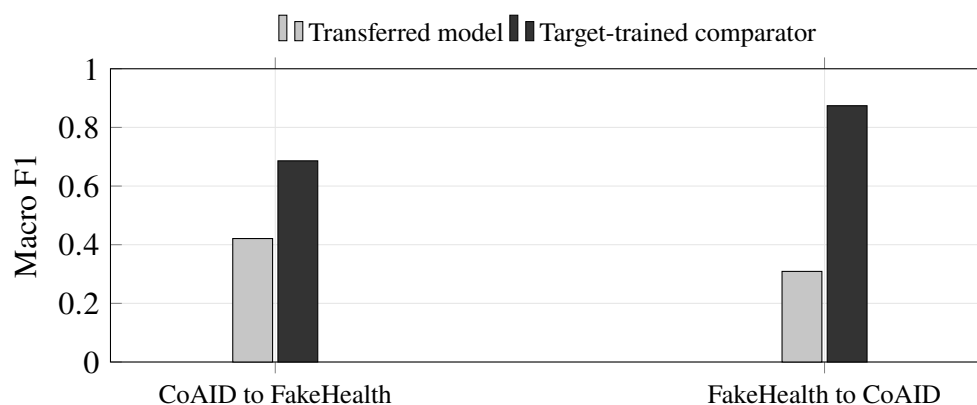


Figure 2

Full-text Logistic Regression transfer versus a target-trained comparator evaluated on the same filtered target records.

Decision-support analyses are useful but do not support autonomous adjudication. Platt-calibrated CoAID Logistic Regression reduces selective risk from 3.9% to 0.5% at 84.5% test coverage, but retained cases are almost entirely reliable-side predictions; unreliable-side coverage is zero. FakeHealth coverage is 81.6%, while risk declines from 30.0% to 24.5%, still too high for automatic health-information decisions. Deferral can prioritize review, not confer medical validity.

On matched FakeHealth records, adding log-transformed engagement counts changes macro F1 from .667 to .677 standard and from .687 to .700 publisher-disjoint. Both paired confidence intervals include zero, so incremental benefit is not established. Narrative discovery finds no size-eligible CoAID cluster. Four FakeHealth clusters undergo human review; three are retained, covering 321 of 720 unreliable-labeled items and describing cancer advances, women's reproductive or menopause treatments, and cardiac procedures or devices. This 44.6% is corpus prevalence, not internet prevalence. The 80-case qualitative audit identifies topic shortcuts and medical-terminology errors frequently within its deliberately enriched sample. Because Stage B proposals were generated and then approved without overrides, these codes are exploratory assisted-review evidence rather than independent agreement data.

Discussion and Implications

Classification performance is conditional on dataset construction. CoAID achieves excellent internal scores while publisher identity remains more predictive than language. FakeHealth’s publisher-disjoint result is more encouraging, but domain uncertainty and truncation remain substantial. Failed bidirectional transfer confirms that these reliability labels do not define a stable universal construct, consistent with warnings about heterogeneous topics and labeling strategies (Baris Schlicht et al., 2023).

For management practice, the appropriate product is review support, not an automated truth engine. It should present risk, confidence, source and temporal context, sensitive terms, narrative, and a defer option. Governance should define expert escalation, retain an audit trail, and monitor publisher, topic, and calibration drift. Source identity must not replace claim-level assessment; engagement volume should inform reach only after exposure and missingness are understood.

Limitations. CoAID is COVID-specific and source-structured; FakeHealth uses health-news review scores. Only 233 CoAID primary articles are unreliable, preventing publisher-disjoint testing. FakeHealth transformers see 512 leading tokens although 81.8% of items are longer. The domain test has few effective groups, neural comparisons use three seeds, engagement is incomplete, and one reviewer completed the qualitative stages. Topics are descriptive clusters, not validated claims. Future exact reruns should seed immediately before random classification-head construction and archive results from a clean tagged revision.

Recommendations. Evaluate contemporary, independently annotated data with trained or clinical reviewers; obtain enough minority examples and publishers for source-disjoint testing; compare models on matched 512-token inputs before testing chunking or longer contexts; seed initialization and archive against a clean commit; and calibrate, monitor, and conservatively defer any operational model. A second reviewer should independently recode qualitative cases.

Conclusion

The project succeeds as a robustness audit. Models discriminate dataset-defined reliability, but strong internal performance coexists with publisher shortcuts, domain sensitivity, and transfer failure. Transformers improve CoAID point estimates without a clear FakeHealth gain; engagement adds no established value. Calibration, narratives, explanations, and deferral instead provide review context. The defensible path is human-centered monitoring with transparent uncertainty, not autonomous medical fact-checking.

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